

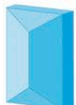


Prisms of rock crystal, a colorless type of quartz

The trigonal crystal system of quartz is visible here.



Orthorhombic
The mineral topaz forms beautiful orthorhombic crystals, often with a pyramid-like top.



Axinite, a silicate mineral (see p.23)

Triclinic
The least symmetrical of all, triclinic crystals include axinite (above) and turquoise.



Ice crystals

In an ice crystal, water molecules are aligned hexagonally. These crystals form when water vapor in the air freezes straight to a solid. If liquid water freezes slowly, it will form simple hexagonal crystals, but without the delicate branches and shapes of a snowflake crystal.



The unique pattern of a snowflake is based on a six-sided shape (hexagon).

Snowflake

A snowflake is a six-sided ice crystal. Each snowflake grows into a different variation on this shape, depending on how it drifts down from the sky. No two snowflakes are the same.

Sugar and salt crystals

Crystals of sea salt and crystals of sugar are more different than they look. Salt crystals are highly ordered six-sided cubes, while sugar crystals are less well ordered hexagonal prisms.

Sea salt belongs to the cubic crystal system, but when the crystals form quickly they take a pyramid shape.

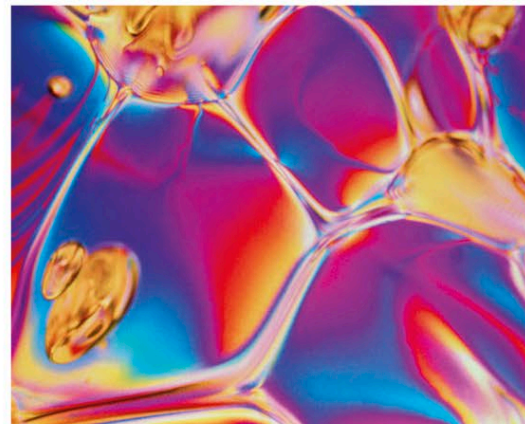


Sea salt crystals

Crystals of sea salt (sodium chloride) are held together by ionic bonds (see p.16). When salt water evaporates, the dissolved minerals left behind form salt crystals.

Liquid crystals

In nature, cell membranes and the solution produced by silkworms to spin their cocoons are liquid crystals. The molecules in liquid crystals are highly ordered, but they flow like a liquid.



Liquid crystals at work

Man-made liquid crystals, such as the ones seen here, are used in liquid crystal displays (LCDs) in TV screens, digital watches, and mobile phones. They do not produce light, but create clear images by altering the way light passes through them.